

CURRICULUM VITAE

Vijay Tiwari

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CAREER OBJECTIVES

"Seeking a challenging teaching role in the department of mechanical engineering that enables me to share my knowledge, skills, and research expertise with students while contributing to the academic community through innovative pedagogy, curriculum development, and scholarly activities."

EDUCATIONAL & PROFESSIONAL CREDENTIALS

● Ph.D. VELLORE, Tamil Nadu.	-	July 2022 (IFT-Pursuing) VIT,
● Master of Technology (Machine Design) GGU Central University, Bilaspur, CG.	-	2013 Passed with 8.5 CGPA
● BACHELOR OF ENGINEERING (MECHANICAL) Maharana Pratap Engineering College, AKTU University, Lucknow.	-	2011 Passed with 69.88 %
● BOARD OF INTERMEDIATE SMLK Nawabganj, KANPUR.	-	2005 Passed with 75.20%
● SECONDARY SCHOOL CERTIFICATE SMLK Nawabganj, KANPUR.	-	2003 Passed with 76%

AREAS OF EXPERTISE

Machinery Fault Diagnosis, Acoustic, Mechanical Vibrations, Condition monitoring, Machine Learning, Deep Learning, Machine Design, Renewable Energy.

PROFILE SUMMARY

- A student-oriented individual with expertise in handling student queries and **7+ years'**

experience as a Lecturer in Mechanical engineering **subjects**. Proficient in developing skills in students and providing them with a growing path.

- Extensive experience in organizing workshops on Technical Skills. Excellent knowledge of the subject and also great practical knowledge. Outstanding track record in assuring student success.
- A clear understanding of **AUTO CAD** tool Concepts.
- Clear Understanding of Solid Modeling, Surface Modeling, Assembly, Drafting & Detailing.
- An effective communicator with exceptional relationship management skills. Ability to relate to people at any business level and significant experience working with Students and colleagues.
- Strong problem-solving and analytical skills, with the flexibility to work cohesively as part of a multidisciplinary team and complement the skills of others.
- A consummate professional with excellent planning and execution skills, attention to detail, and the ability to build and lead effective teams.
- Coordination with R&D for Technical requirements. Complete the performance with competitors for better performance. Renovate new concept designs to improve the performance of existing models.

WORK EXPERIENCE

Designation	College/Industry	Work Period
Assistant Professor	Parthivi college of Engineering & Management Bhilai, Durg	14-01-2014 to 06-06-2014
Assistant Professor	Dr. Ambedkar Institute of Technology for Handicapped, Kanpur	22-8-2015 to 14-06-2019
Assistant Professor	Dr. Virendra Swarup Group of Institution, Unnao	06-08-2019 to 30-07-2022

TRAINING, RESEARCH AND DEVELOPMENT ACCOUNTABILITY

- Finished Project, principal Investigator, "Electromagnetic Breaking System" at MPEC, Kanpur, India.
- Finished Project, principal Investigator, "Design of Sliding Mode Controller Enhanced by Fuzzy Logic Algorithm for Industrial Robot" at GGU University, Bilaspur, Chhattisgarh.

ADMINISTRATION

Designation	College/Industry	Work Period
Assistant Professor	Dr. Virendra Swarup Group of Institution, Unnao	06-08-2019 to 30-7-2022
HoD	Dr. Virendra Swarup Group of Institution, Unnao	26-09-2020 to 16-10-2021

KEY RESPONSIBILITIES HANDLED

- Manage the teaching plan for students and provide them with practical knowledge.

- Identify the weak points of students.
- Organize various events to enhance the skills of students.
- Supervising the abilities of students.

ENGG SUBJECTS THAT I CAN HANDLE

- Neural Networks and Deep Learning.
- CAD/CAM.
- Production planning and control.
- Metrology, QC & Reliability.
- Engineering Drawing.
- Production drawing.
- Machinery Fault Diagnosis and signal Processing
- Mechanics of Machinery.
- Mechanical Vibrations.
- Rotor Dynamics.
- Noise Control & Management
- Strength of Materials.
- Fluid mechanics.
- Engineering mechanics.
- Sensing and Measuring Principle
- Design of machine elements.
- Signal Processing and Its application.

❖ Ability to conduct Technical Labs, such as **Design and Analysis** Domains (**MATLAB, AUTO CAD**)

NPTEL Course Completed /Ongoing

- Machinery Fault Diagnosis and Signal Processing Jan-Apr 2023
- Data Science for Engineers Jan-Mar 2024
- Data Analytics with Python Jan-Apr 2024
- Noise Management & Control Jan-Apr 2025
- Wind Energy Jan-Apr 2025

TECHNICAL PROFICIENCY

Programming Languages	:	Python, R Programming
Operating System	:	Windows XP/2003/2008, Vista, Win7, Windows Server 2003, Windows Server 2008 & Win7 Server, Windows 10 & 11.
Software	:	MS Office, LaTeX.
Design Tools	:	AUTO CAD.
Analysis Tools	:	ANSYS 12.0, MATLAB.

FACULTY DEVELOPMENT PROGRAMME/WORKSHOP

1. Attended “The SERB Sponsored One Week High-End Workshop on “**Artificial Intelligence Techniques for Machinery Condition Monitoring**” (using MATLAB and LabVIEW) (AITMCM-2023), organized by the Department of Mechanical Engineering, NIT Rourkela, from 9th to 15th January 2023.
1. Attended ATAL FDP " **Sensors and their applications**", organized by **Department of Electronics engineering at ST. MOTHER THERESA ENGG COLLEGE** from **17/02/2025 to 22/02/2025**
2. Attended “**Vibration Acoustic and control in industrial Engineering**” organized by **Department of Mechanical**

engineering at ISM Dhanbad from 9/04/24 to 12/04/2024

3. Attended **ATAL FDP “Green Technology towards Sustainable Future”** at **LAXMINARAYAN INNOVATION TECHNOLOGICAL UNIVERSITY** from **03/02/2025 to 08/02/2025**.
 4. Attended **“Machine Learning: Next Generation Intelligent E-Mobility”** organized by **IEEE Power Electronics Society Harcourt Butler Technical University (HBTUK), Department of Electrical Engineering (HBTUK)** on **23rd September, 2023**.
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INTERNATIONAL JOURNALS

1. Vijay Tiwari, Lalit Kumar Bansal,” Fault Detection of Wind Turbine Blades: A Hybrid CNN-LSTM based Approach using Acoustic Signature Analysis”, Neural Computing and Applications 2025, Springer (**under Review, IF4.5**).
2. Vijay Tiwari, Lalit Kumar Bansal,” Acoustic Based Condition Monitoring of Wind Turbine Blades: A Comparative Study of Machine Learning Approaches”, Measurement 2025, Elsevier (**under Review IF 5.2**).
3. Kailash Kumar Borkar, Piyush Tiwari, Vijay Tiwari, “Heuristic Rule Base Network Path Analysis and Planning of Wheeled Mobile Robots”, International Journal of research in Aeronautical and Mechanical Engineering, ISSN -2321-3051, vol.1, Issue 7, nov.2013, pp. 54-58.
4. Piyush Tiwari, Kailash Kumar Borkar, Vijay Tiwari, “Efficient Approach for Minimizing Tool Paths Length for Bezier Surface from Iso-Scallop Technique”, International Journal of research in Aeronautical and Mechanical Engineering, ISSN -2321-3051, vol.1, Issue 7, nov.2013, pp.47-53.
5. Vijay Tiwari, Kailash Kumar Borkar, Piyush Tiwari, “Design of Sliding Mode Controller Enhanced by Fuzzy Logic Algorithm for Industrial Robot”, International Journal of research in Aeronautical and Mechanical Engineering, ISSN -2321-3051, vol.1, Issue 7, Nov., pp.144-148.

INTERNATIONAL CONFERENCE PUBLISHED/Proceeding

- 1.**Vijay Tiwari, Lalit Kumar Bansal,**” Efficient Condition Monitoring of Small-Scale Wind Turbine Model Through Supervised Machine Learning Classification Using Acoustic Data”, Energy Center, MANIT, Bhopal,21-22Feb 2025.
- 2.**Vijay Tiwari, Kailash Kumar Borkar, Piyush Tiwari,** “Heuristic Rule Base Path Planning and Path Optimization of Non-Holonomic Wheeled Mobile Robots”, International Conference on Advanced Research Applications in Engineering and Technology, Rungta Engineering College, Raipur, ISSN -2321-3051, 23-24 March 2014.
- 3.**Vijay Tiwari, Kailash Kumar Borkar, Piyush Tiwari,** “Design Adaptive Nonlinear Noise Cancellation Using the Fuzzy Logic for Robots”, International Conference on Advanced Research Applications in Engineering and Technology, Rungta Engineering College, Raipur, ISSN -2321-3051, 23-24 March 2014.
4. **Vijay Tiwari, Kailash Kumar Borkar, Piyush Tiwari, M.K. Singh** “Path Optimization of Non- Holonomic wheeled Mobile Robots Using HRBN Technique”, International Conference on Emerging Trends in Electrical, Electronic Instrumentation & Computer Engineering (ETEICE), Bhilai Institute of Technology, Raipur, ISSN Print-2321-5747, 27-28 March 2014.
5. **Vijay Tiwari, Kailash Kumar Borkar, Piyush Tiwari, M.K. Singh** “Design a Neural Network controller for Noise Cancellation during Trajectory Tracking for Manipulator Using MATLAB”, International Conference on Emerging Trends in Electrical, Electronic Instrumentation & Computer Engineering (ETEICE), Bhilai Institute of Technology, Raipur, ISSN Print-2321-5747, 27-28 March 2014.
- 6.**Vijay Tiwari, Kailash Kumar Borkar, Piyush Tiwari, M.K. Singh,** “Degree reduction of Bezier patches for sculptured surface: A Review”, International Conference on Emerging Trends in Electrical, Electronic Instrumentation & Computer

AWARDS/RECOGNITIONS

1. Gate score 397 in Graduate Aptitude Test in Engineering 2011.
 2. Worked as a Mentor in Induction Programme at Dr. AITH Kanpur.
 3. Associate Member of IEI with membership no. is AM175605-5
 4. Organized the event Auto CAD QUIZ in “YANTRIKI-2009” organized by “Department of Mechanical Engineering.
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PROJECTS- UG STUDENTS

- Design of Modified Wheel Chair (2015-16).
 - Design and Optimization of Electric Motor Driven Mechanical Press Type Seed Oil Extraction Machine (2016-17).
 - Design and Manufacturing of Four-Way Hacksaw Machine (2017-18).
 - Design and Analysis of Differential Gear Box (2017-18).
 - Design and Analysis of Hydraulic Ram Pump (2017-18).
 - Design and Modelling Multipurpose Operation Drilling, Hacksaw, Cutting and Grinding Machine (2017-18).
 - To study the microstructural and mechanical properties of aa6061 aluminum alloy processed by Equi channel Angular rolling (e-car) (2019-20).
 - Mechanical characteristics of jute, chopped e-glass reinforcement epoxy polymer composite (2019-20).
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TRAINING AND CERTIFICATION TOOLS

Year

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|---|---------|
| • Auto Cad from CADD CENTRE | 2010-11 |
| • Taken 1 days of training at VIT Vellore for CREO. | 2011 |
| • Trained on JMP PRO Data analysis Tool | 2023 |
| | &24 |
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MASTERS Academic Project

PROJECT # Design of Sliding Mode Controller Enhanced by Fuzzy Logic Algorithm for Industrial Robot.

Duration : 2012-2013
Tools used : MATLAB R2009a,
Description :

- ❖ SMC is a robust control method designed to handle system uncertainties and external disturbances. FLA introduces adaptability by using fuzzy rules to approximate nonlinearities and uncertainties in the system. The enhanced controller can achieve better tracking performance, stability, and robustness. The stability of the control system is often proven using the Lyapunov stability method, and performance metrics like RMS error are used to evaluate the effectiveness of the controller.
 - ❖ In this project, the robot arm dynamic aspects of the system, such as Proper rotor dynamic analysis of manipulator configuration, joints of link, End-effector, Actuators, Sensors, controller were described. The theoretical aspects and software available and various aspects of design and chattering noise of arm manipulator were studied in detail.
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POSITIVE STRENGTHS

- Quality conscious, interested in learning and working on new technologies & Subjects.
 - Positive attitude and Practical approach.
 - Good grasping & observing power.
 - Self-driven with high commitments.
 - Interpersonal & Communication Skills.
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PERSONAL DETAILS

Name	:	Vijay Tiwari
Fathers Name	:	Sri. Shokhi Lal Tiwari
Mother Name	:	Smt. Premvati Tiwari
Date of Birth	:	Oct 12 th , 1988
Sex	:	Male
Marital Status	:	Married
Nationality	:	Indian
Hobbies & Interest	:	Reading Books, Playing Cricket
Languages known	:	English, Hindi
Contact Address	:	C/O Shokhi Lal Tiwari, House No:33/203, Near Suryansh Hospital, Balaji Puram, Makdi Kheda Indra Nagar Road Kalyanpur Kanpur-208017
References	:	Available on Request

PERSONS WHO ARE FAMILIAR WITH THE APPLICANT'S WORK

- **Dr. Ganesh Prasad Shukla**, Associate Professor, Industrial & Production Engineering dept., G G U University, Bilaspur, India.
Ph: +91- 8889254445. Email Id: ganeshshukla_2006@yahoo.com
 - **Dr Lalit Kumar Bansal**, Associate Professor Grade 1, Department of Thermal Engineering and Energy, School of Mechanical Engineering, VIT, Vellore, India.
Ph: +91- 99451 67343 Email Id: lalit.bansal@vit.ac.in
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DECLARATION

I am familiar with Information Technology aspects and am confident in my ability to work in a team. I now declare that the information furnished above is accurate to the best of my knowledge.

Date:

Place:

A handwritten signature in black ink, appearing to read 'Vijay', with a horizontal line drawn through the middle of the letters.

VIJAY TIWARI)